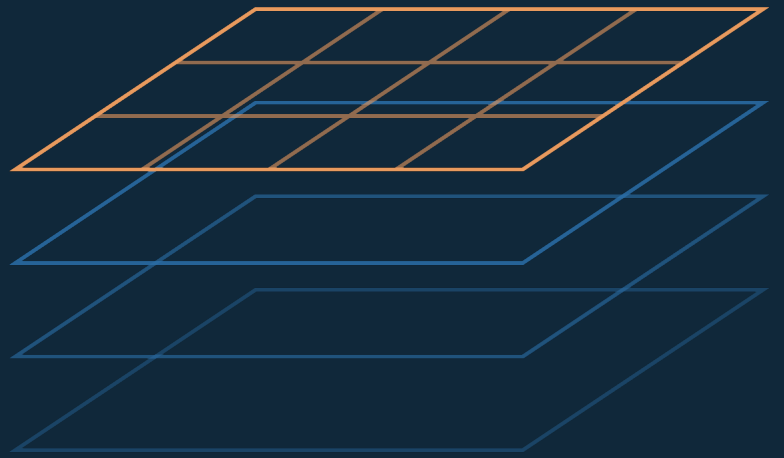


LOCMARCA 2026 · COURSE 0a

Anatomy of a netCDF file

What is inside the files you will open all week, and how to read them

LOCMARCA 2026

INTRODUCTION

Why not a spreadsheet?

Self-describing

Data and their description (units, origin, grid) travel in the same file

Many dimensions

Natural for fields that depend on time, depth, latitude and longitude

Binary, compact

Fast to read, and you can read only the part you need

A standard

Read by Python, R, Matlab, CDO, NCO, ncview… on any computer

22 million lines: what one year of daily SST on our small Costa Rica grid would need as a CSV table (365 days × 200 × 300 points).

THE ONE IDEA

A netCDF file is a labelled box: a header that describes the content, and the data themselves.

You can read the label without opening the box: **ncdump -h** reads only the header, instantly, even for a file of several GB.

BUILDING BLOCKS

Four building blocks

1 · Dimensions

The named axes and their sizes

```
time = 365 ;
```

2 · Coordinate variables

The values along each axis

```
latitude = 4.025, …, 13.975 ;
```

3 · Data variables

The physical fields, defined on the dimensions

```
analysed_sst(time, latitude, longitude)
```

4 · Attributes

Metadata attached to a variable or to the whole file

```
analysed_sst:units = "kelvin" ;
```

Everything else is a combination of these four. In xarray: **ds.sizes**, **ds.coords**, **ds.data_vars**, **.attrs**.

BUILDING BLOCKS

Reading a real header: ncdump -h

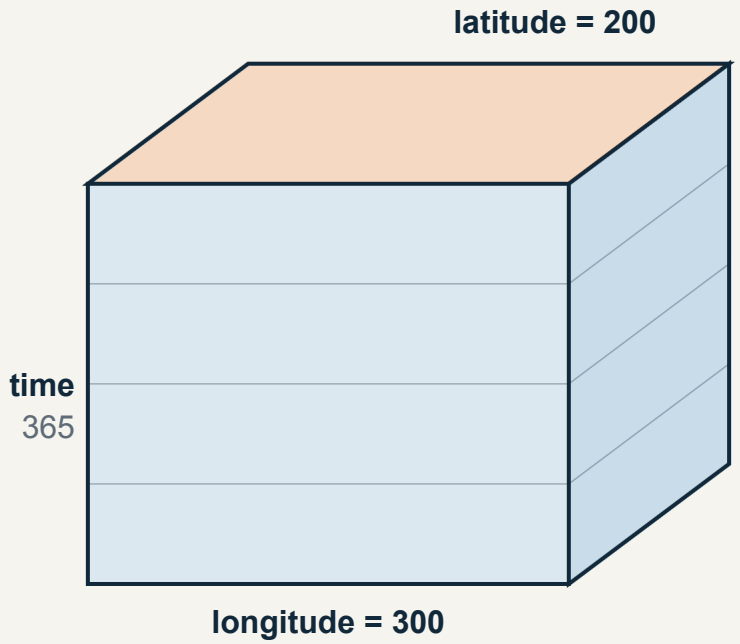
```
netcdf sst_costarica_2023 {
dimensions:
    time = 365 ;
    latitude = 200 ;
    longitude = 300 ;
variables:
    short analysed_sst(time, latitude, longitude) ;
      analysed_sst:long_name = "analysed sea surface…" ;
      analysed_sst:units = "kelvin" ;
      analysed_sst:scale_factor = 0.01 ;
      analysed_sst:add_offset = 273.15 ;
      analysed_sst:_FillValue = -32768s ;
    float latitude(latitude) ;
    float longitude(longitude) ;
    int time(time) ;
      time:units = "seconds since 1981-01-01" ;
```

- **Dimensions**: names and sizes
- **Data variable**: type, name, dimensions
- **Coordinate variables**: same name as their dimension
- **Attributes**: variable:name = value

The header of sst_costarica_2023.nc (TP1b), shortened.

BUILDING BLOCKS

Dimensions give the shape of the data



Size: 365 × 200 × 300 = **21.9 million** values. Stored as 2-byte integers, that is ≈ 44 MB: the size of the file.

Order matters: analysed_sst(time, latitude, longitude). Each time step is one map.

Names are free: WOA calls its dimensions **X, Y, Z, T** (360 × 180 × 33 × 4). Always read them first.

BUILDING BLOCKS

Coordinates turn an index into a place

latitude	[0]	[1]	[2]	…	[199]
value (°N)	4.025	4.075	4.125	…	13.975

Index or value

isel(latitude=0) takes position 0; **sel(latitude=4.025)** takes the value. Same point.

Spacing

0.05° between two values: the grid spacing (≈ 5.5 km), not the effective resolution (Course 1, tomorrow).

Direction

Here south to north. Some files store north to south: check before any slice.

BUILDING BLOCKS

One file, several variables

```
dimensions: depth = 31 ; latitude = 121 ; longitude = 181 ; time = 31 ;
short thetao(time, depth, latitude, longitude) ;  // sea_water_potential_temperature
short so(time, depth, latitude, longitude) ;      // sea_water_salinity
short uo(time, depth, latitude, longitude) ;      // eastward_sea_water_velocity
short vo(time, depth, latitude, longitude) ;      // northward_sea_water_velocity
```

One month of the GLORYS12 reanalysis (TP2a, TP3): **4 variables** share the same 4 dimensions.

Short names (**thetao**, **so**…) change from producer to producer: the attributes tell you what they are.

Type	Bytes	Typical use
short	2	Packed data (next slides)
int	4	Counts, time
float	4	Most fields, coordinates
double	8	High precision, time

ATTRIBUTES

The attributes you must read

Attribute	In the OSTIA file	Why it matters
units	kelvin	300 K is 27 °C: convert before plotting
long_name	analysed sea surface temperature	Human-readable description
standard_name	sea_surface_foundation_temperature	Exact meaning, from the CF list
_FillValue	-32768	Marks missing points (land)
scale_factor, add_offset	0.01, 273.15	Unpack stored integers into real values
valid_min, valid_max	-300, 4500 (packed)	Values outside are not physical

ATTRIBUTES

Packing: integers that hide real numbers

value = stored × scale_factor + add_offset

```
2811 × 0.01 + 273.15 = 301.26 K = 28.11 °C
```

Why

2 bytes instead of 4 or 8: the file is 2 to 4 times smaller, at 0.01 K precision.

Good news

xarray, CDO and ncview unpack automatically: you see 301.26.

Trap

ncdump -v prints the raw integers: 2811, not 301.26.

ATTRIBUTES

Missing values: one number means "nothing"

File	Attribute	Value	Where
OSTIA SST	_FillValue	-32768	Land
WOA 2009	missing_value	-99.9999	Land and below the sea floor

When the tool knows

The value becomes **NaN** (Not a Number): blank on maps, skipped by means.

When it does not

-99.99 °C enters your average. Always check the **min and max** of a new variable.

ATTRIBUTES

Time is stored as a number

date = reference date + value × unit

File	time:units	Example
OSTIA SST	seconds since 1981-01-01	1325376000 → 2023-01-01
GLORYS12 reanalysis	hours since 1950-01-01	Another reference, another unit
WOA 2009	months (no reference)	1.5, 4.5, 7.5, 10.5: mid-season, not dates
CROCO model (TP3)	seconds (no reference date)	Start of the model run: you add it (TP3)

Also read **calendar**: climate models often use 365-day years (no 29 February) or 360-day years (Course 2).

GLOBAL METADATA

Global attributes: the file's identity card

Attribute	In the OSTIA file
title	Global SST & Sea Ice Analysis, L4 OSTIA, 0.05 deg daily
institution	UKMO (the UK Met Office)
source	AMSR2, GOES, SEVIRI, SLSTR, AVHRR, VIIRS… (the satellites merged)
Conventions	CF-1.4, ACDD-1.3
history	Created from sst.nc; obs_anal.nc; seaice.nc
acknowledgment	How to cite the data in your report or paper

history grows: NCO and CDO append each command you run (TP1b). Your own processing leaves a trace.

GLOBAL METADATA

CF conventions: a shared vocabulary

The **Climate and Forecast (CF)** conventions say how to write units, axes, time and variable names, so that tools understand a file without help.

What the tool looks for	OSTIA (CF-1.4)	WOA 2009 (no CF declared)
standard_name	On every variable	None
axis (X, Y, T)	On every coordinate	None: only the names X, Y, Z, T
Time units	seconds since 1981-01-01	months: no reference, not decoded
Missing values	_FillValue	missing_value

A CF file "just works" in xarray, CDO, ncview. With WOA, **you** must tell the tools which dimension is which (TP1b).

GLOBAL METADATA

Metadata can be wrong

Three real attributes of sst_costarica_2023.nc, a 2023 file covering 92°W–77°W:

`time_coverage_start = "19811001T000000Z"`	The data start on 1 January 2023
`easternmost_longitude = 180.00002`	The grid stops at 77°W
`analysed_sst:coordinates = "lon lat"`	The variables are latitude and longitude

Cutting a sub-region copied the attributes of the global product. **Trust the coordinates and the data; check the attributes.**

TOOLS

Six ways to look inside

Tool	Command	What you get
ncdump	ncdump -h file.nc	The header only, instantly
ncdump	ncdump -v time file.nc	Header plus the values of one variable (raw, packed)
NCO	ncks -m file.nc	The metadata
CDO	cdo sinfo file.nc	A summary: grid, levels, time steps
ncview	ncview file.nc	Interactive quick-look maps (needs a graphical display)
Python	xr.open_dataset("file.nc")	Decoded: real dates, NaN, unpacked values

Same file, different views. In a Jupyter cell, prefix shell commands with **!** : !ncdump -h {SST_FILE}

PREVIEW OF COURSE 1

The five questions, answered by the header

Question	Where to look
1 · What exactly is measured?	standard_name, long_name, units
2 · Where and when?	Range of the coordinate variables; time:units and the time values
3 · Which resolution?	Spacing of the coordinates (here 0.05°): the grid, not the effective resolution
4 · Climatology or real period?	Number and meaning of the time steps: 4 seasons (WOA) or 365 days (OSTIA)
5 · How uncertain?	Error or quality variables (e.g. analysis_error), references, documentation

The header answers the first four in two minutes. The fifth often needs the product documentation.

NEXT

Your turn: TP1a, then TP1b

1

Open

TP1a, section 1: the OSTIA file with xarray, the four blocks

2

Find

TP1b, B.1 and B.2: the same header with ncdump -h and cdo sinfo

3

Check

Units, time range, latitude order: before any map or average

Questions?